Assessed Coursework Cover Sheet for PSYC0101

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MODULE NAME & CODE: PSYC0101: INTERACTION SCIENCE
COURSEWORK PART (IF RELEVANT): PART 1
NAME OF TEACHER WHO SET THIS WORK: DUNCAN BRUMBY
COURSEWORK TITLE: UNDERSTANDING EXISTING HCI RESEARCH
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Coursework 1 – Part A: Describing Existing HCI Research

Paper Summary 1

People interact with various types of devices and interfaces frequently in their daily life. The frequency of these interactions could lead to habit formation. Despite an emerging popularity of habit research in the field of HCI, there is currently little quantitative research on how habit development and disruption affect user performance. The aim of Garaialde et al. (2020) was to provide a quantitative study of the effect of habit formation and disruption on user performance. A lab study and a study in the wild are conducted to examine how building and breaking a habit affect the speed and accuracy of interactions.

Both studies adopted within-subject design. The independent variable was experiment phase (baseline, habit, disruption); The dependent variables were speed and accuracy. All images used were courtesy of Snodgrass and Vanderwart (1980).

In the lab study (study 1), a laptop using Psychopy v1.8 was used as display. 19 (12 men, 7 women) participants were asked to make a choice of whether the correct object label was on the top-left or top-right corner of the screen. In the baseline phase, the correct label appeared randomly on either left or right; In the habit phase, the correct label remained static; In the disruption phase, it became random again. Participants were not made aware of the change beforehand. Each image was made to appear equal amount of time in each phase.

The field study (study 2) was run on Android phones of participants. 18 (8 men, 10 women) participants were asked to pick the right image specified by the text notification from two images vertically displayed. Experiment was divided into 3 phases and the correct image appeared statically in habit phase, randomly in the rest. Time duration of each phase was ensured to be at least 22 days. The entire study lasted for 78 days.

Study 1 revealed that habit formation leads to speed increase, disruption of habits leads to speed decrease, habit formation results in accuracy increase, and disruption of habits results in seed decrease. Notably, when habit was disrupted, any of the performance improvements were found erased. The finding of habit-speed relation in study 1 was supported study 2. However, contrary to the finding of habit-accuracy relation in study 1, study 2 showed there was no statistically significant change of accuracy after habit formation and disruption. Overall, user response times were longer in the wild.

This research provided quantitative evidence how habit formation and disruption affect user performance. It emphasised the importance of supporting habit formation during design and preventing habit disruption during design changes, as well as the need of purposeful habit breaking in certain circumstances.

Word count: 436/400

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Garaialde, D., Bowers, C. P., Pinder, C., Shah, P., Parashar, S., Clark, L., & Cowan, B. R. (2020). Quantifying the impact of making and breaking interface habits. *International Journal of Human-Computer Studies*, *142*, 102461. https://doi.org/10.1016/j.ijhcs.2020.102461

Snodgrass, J. G., & Vanderwart, M. (1980). A standardized set of 260 pictures: Norms for name agreement, image agreement, familiarity, and visual complexity. *Journal of Experimental Psychology: Human Learning and Memory*, *6*(2), 174–215. https://doi.org/10.1037/0278-7393.6.2.174

Paper Summary 2

Human multitasking involves regular interleave attention between multiple concurrent tasks. Given the prevalence of multitasking and its safety and productivity implications, it is vital to understand how well people deal with the issue of attention interleaving. Yet previous studies fail to define the priority criterion objectively. The aim of Farmer et al (2017) was to understand how priorities, formalized through a payoff function, affect attention allocation during multitasking. They investigated how people decide to interleave two concurrent tasks and whether their dual-task performances are optimal.

Experiment 1 adopts a 3 x 2 (payoff function x cursor noise) within-subjects design. 20 university students (13 male) were recruited. Participants were required to perform a typing task where a list of digits presented were to be typed to gain 10 points (when typed correctly) or lose 5 points (when typed incorrectly) and a tracking task where a joystick is used to keep the moving cursor inside the target area to prevent penalty. The cursor dislocates regularly in either of 2 types of noises (3 or 5 pixels standard deviation). The penalty magnitude is determined by one of the 3 payoff functions (lose 500 points, lose all, lose half). Each dual-task trial lasted for 120s.

Experiment 2 has three phases, exploration, exploitation and transfer. In the first 2 phases, payoff function “lose 500” and “lose half” were tested in a between-subjects study. In the transfer phase all participants experienced payoff function “lose 200%”. 13 university students (7 female) were recruited. The task and materials remained the same as experiment 1, except the cursor movement had a standard deviation of 4 pixels and participants now earn 4 points or lose 1 point from typing correctly or incorrectly. Each trial lasted for 30s.

Through ANOVA, results from experiment 1 show that participants are capable of strategically adjust their time spent on one task before task-switch, according to the current payoff structure and task difficulty. And the strategies are often close to but not at optimal as predicted by an optimal performance model, which indicates that participants were somewhat risk averse. Results from experiment 2 further evidenced findings in experiment 1 and suggest that participants may transfer strategies from one payoff system to another. Participants who experience a harsher payoff function were less likely to change their strategy boldly in face of a new payoff function, vice versa.

This study has provided detailed empirical and quantitative analysis of how people adapt to performance explicit feedback when multitasking, as well as addressed people’s flexibility in strategy change upon condition change and its limitations, offering vital information for the cognitive science research and multitasking system design.

Word count: 437/400

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Farmer, G. D., Janssen, C. P., Nguyen, A. T., & Brumby, D. P. (2017). Dividing Attention Between Tasks: Testing Whether Explicit Payoff Functions Elicit Optimal Dual-Task Performance. *Cognitive Science*, *42*(3), 820–849. https://doi.org/10.1111/cogs.12513

Paper Summary 3

A self-interruption is when a user voluntarily switches a way from a task before it is completed. Studies on workplace multitasking showed that every 12 minutes people are interrupted, and almost 50% of them are self-initiated (Mark et al, 2005). However, as most of the research focuses on external interruptions, internal interruptions (self-interruptions) are yet not well understood. Jin and Dabbish (2009) wanted to examine this type of task-switching behaviour in detail and tried to understand the following: The reason people self-interrupt on computer and switch to some other tasks, the characteristics of internally generated interruptions, the potential positive and negative side-effects of self-interruption. Through a qualitative study they aim to introduce a new typology on self-interruption on computer through a qualitative study.

With a grounded theory approach, the authors generated a descriptive theory of the cause and effects of self-interruption. Data was obtained through observational studies and a retrospective interview with each participant on their motivation for task-switch. 13 participants (6 men, 7 women) were selected based on their work and availability from around a local university. They were shadowed by an observer for approximately an hour doing their own work on the computer in their workplace as usual. During the observation, the researcher would take note of instances when the participant switch between virtual or physical activities or environments. The researcher would note the time, activity preceding and following the switch. No distinction between external or internal interruption was made.

A 30 minute to 1 hour retrospective interview was followed immediately after the observation. For each task switch observed, participants were asked about the pre-switch activity, post-switch activity and the reason for switching.

Theoretical sampling method was utilised to identify and analyse the internal interruptions in the data collected. A new dataset containing each self-interruption observed, along with the broader work context, activities preceding and following the switch, interruption duration and the motivation was created and analysed using open and axial coding techniques for categorisation.

Self-interruptions were divided into 7 typologies using the grounded theory: adjustments, breaks, routines, waits, inquiry, triggers and recollections. Each type has its unique properties. Design ideas were suggested to address their negative consequences if any. While considering the similarities and differences across different types of self-interruption,

3 main themes were discovered: Whether the interruption is self-motivated or situationally motivated, whether it is stimulus based or goal-oriented shift in focus of attention, and whether it creates a new task.

This study has greatly extended the understanding of self-interruption on the computer and contributes to the literature on attention and interruption. It could also inform future designs of interactive systems intend to support digital multitasking.

Word count: 440/400

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Mark, G., Gonzalez, V. M., & Harris, J. (2005). No task left behind? *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. https://doi.org/10.1145/1054972.1055017

Total word count for Part A: 1313 / 1200

Coursework 1 – Part B: Applying Existing HCI Research

Introduction

User instruction, a concept that has accompanied intricate human creations in various forms, has long been a positive force for the spread of knowledge. First user instruction is believed to date back to 205 BC (Carman et al., 2014), engraved on the cover of a Greek analogue computer (Efstathiou et al., 2018). The publication of Maxon's Mechanick Excercise (Maxon, 1683) recognised as the first comprehensive manual on printing, is often treated as a pioneering systematic guideline of modern machinery (Carpenter, 1959). In today's consumer electronics industry, a paper manual that comes with the product is often taken for granted. Countries have even established regulations to mandate the inclusion of manuals, mostly for consumer electronics (e.g., ISO 20607: Instruction Handbooks for Machinery).

However, as we enter the information age, product designers have discovered unprecedented ways to implement the concept of user instruction. Users' expectation of instructions is also affected by the ease of access to information. The necessity of a paper-format user manual has become debatable. Do we not need a printed user manual anymore? From a product designer's perspective, the answer should be case specific. There is no doubt, however, for a specific product, there is a definitive answer. In the case of AI (artificial intelligence) home thermostats (or AI thermostats, smart thermostats, used interchangeably), research findings support the opinion that a printed user manual is not necessary. This paper aims to illustrate this argument in detail.

Evidence will be provided to support the argument in three layers. Firstly, general dissatisfaction with the existing instruction system and its cause will be presented. Secondly, a case-specific analysis will show the redundancy of a printed user manual for an AI thermostat. Lastly, a series of superior methods to support user experience will be proposed to stress the need of replacing printed user manuals for AI thermostats.

The Unsatisfied User

There has always been a negative notion towards user manuals among both consumers and designers. A qualitative study run by Schriver (1997) suggests that the younger generation might have a lower frequency of referring to instructions when operating an unfamiliar system. Another similar study carried out by Loggem (2014) despite arguing that people still recognise the necessity of user instruction, revealed a tendency among users to search for information from "unauthorised" sources (e.g., friends and the internet) instead of sticking to the official instruction, and supported the opinion that younger people are less fond of formal instructions. Specifically, this study shows a clear preference for non-documentation information over documentation information, while facing obstacles operating software. This indicates that people prefer seeking assistance outside the user manual despite its legitimacy.

To extend the understanding of users' interaction and opinion towards user manuals across different product realms (e.g., consumer electronics, kitchenware, software, public service), Blackler et al, (2014) conducted a quantitative on 107 participants of different age groups and reported a very small user frequency (<25%). Another longitudinal study over 7 years run by Blackler et al., (2014) informed a general negative emotion of users towards the user manual. In fact, users are reported to expect the product interface to be self-evident, so they do not need to use the manual. Over-featuring of products was reported to be another major problem that affects user emotion towards manuals, as it leads to an excessive amount of information and learning fatigue.

In a task-oriented problem-solving scenario, users would tend to refer to the documentation in combination with trial and error (Rieman, 1996). In the case of first-time user experience, this theory posits that to understand a product, a user tends to start addressing the primary need (e.g., booting up the system). Instead of reading from start to end, a user is likely to perform the first action immediately after learning about it, and start to explore without guidance (i.e., through trial and error). The natural tendency might explain why users' perceptions become particularly negative when the product is over-featured, as they are forced to bounce between action and reading frequently. It also explains why users lean towards "unauthorised" information (Loggem, 2014).

Why Remove the Manual – A Case-specific Discussion

To turn down the need to include user manuals for AI thermostats, the difference between an AI and a conventional thermostat's user interaction should be considered. Although a user will expect an AI thermostat the have the same functions as a conventional one, with the support of IoT (Internet of Things), Machine Learning and Digital Systems technology, its interface is likely to be minimal, and digital. It might even exist outside the physical product (e.g., voice interface, mobile app and automated system). It is also evolving at a fast pace. For instance, Ponce et al., (2019) introduced a consumer-product and product-product social communication framework for smart thermostats that would drastically change Human-machine Interaction. A similar possibility was suggested by the study of Arora et al., (2019), who suggested that IoT and Artificial Intelligence could simplify user interaction by empowering product-product communication. This important trend would impact an inexperienced user in two ways.

Firstly, the first-time user experience will be confusing. Due to the drastic difference between traditional thermostats and AI thermostats, the inexperienced user will realise their old knowledge of thermostats cannot be applied to understanding the basic functions of this AI thermostat at all. Hence, they will try to refer to some kind of instruction. Although a printed manual could address the need at this point, it is not preferred. As discovered by Loggem (2014), the user is more likely to search for "unauthorised" help, predominantly online. Therefore, search-engine-optimised web-based instruction might be more helpful than a printed manual. This will be detailly discussed in the next section.

Secondly, after a user understands the basic function (e.g., turning on and setting the temperature), as posited by Rieman (1996), they will stop seeking help and carry on exploring the other functions until they bump into another obstacle. Because a smart

interface is designed to be responsive, there are two possible outcomes: The user will either by chance build a correct understanding through exploratory learning and keep operating the system correctly, or a wrong understanding but still being able to operate, leading to discouraging outcomes. The latter phenomenon is explained by Nielsen (1998) as an Assimilation Paradox where a new user makes wrong inferences based on existing knowledge. In neither of the situation, a printed manual is useful as there is no powerful prompt that pushes the user back to refer to the official instruction.

In terms of the experienced user's interaction, particularly a user that switches from one smart thermostat to a different one, the negative transfer problem of the Assimilation Paradox (Nielsen, 1998) informed the risk of learning opportunity inhibition, where the user's rich understanding of the prior product obstructs the generation of new inferences. If the user can comfortably apply the old knowledge to the new thermostat, though inefficiently, they will not choose to adjust their knowledge to optimise performance. Therefore, these users' appreciation of instructions is extremely low.

Since user manuals are often associated with annoyance (Blackler et al., 2014), and the usability of a documented instruction for an AI thermostat is generally low, actions should be taken to improve this situation. Removing the printed manual is the first and necessary step and is expected to have a positive impact on user experience.

Due to the physical simplicity of AI thermostats, users are not cued to seek explanations of many (if any) physical features. This naturally eliminates the over-featuring problem (Blackler et al., 2014), and leads to either a minimal manual or no manual at all. Notably, an AI thermostat often comes with a digital display. Positioning instructions onto the display is more intuitive than putting them on a separate booklet. A digitally displayed instruction also has the advantage of adaptively assisting the process of sensemaking, as it aligns with the iterative nature of the Information Journey (Blandford and Attfield, 2010) or the hierarchical characteristics of knowledge acquisition (da Fontoura Costa, 2006), both illustrates the progressive nature of learning that can hardly be attended by non-interactive instructions.

By studying the process of computer-naive users learning about a simple programmable toy with zero guidance, Shrager et al. (1986) demonstrated that it is even possible for inexperienced users to learn a system without any instruction under certain prerequisites. Indeed, the manual is not always necessary. The next section presents these prerequisites for an AI thermostat.

Better Alternatives

With the popularised application of various technologies (e.g., Artificial Intelligence), documentation user guidelines should be replaced by a smarter guidance system. Importantly, this paper is not advocating the elimination of instructions but only proving the redundancy of a printed manual for AI thermostats. When certain prerequisites are fulfilled through the advancement of technology and design, instructions will exist more subtly to continuously help users. After all, the whole point of having a user manual is to prevent user frustration.

A correct mental model is necessary to start a user journey (Kieras et al., 1984; Ritter et al., 2014), hence the first-time user experience is vital. As mentioned in the previous section, because of the huge aesthetic difference between a traditional thermostat and a smart one, users will be cued to search for information through conventional means. Designers should cease the chance to implement the correct "first impression" for the users. For example, basic knowledge could be shown when the device is first turned on, or simply printed on a protection film.

After the first-time experience, heuristic learning should be encouraged. Carroll (1987) suggested that an inexperienced user should be taken as an expert in another field, emphasising the power of inference. Therefore, the digital interface should borrow visuals and patterns from areas that general users are familiar with. For example, a homepage that resembles weather forecasting mobile apps could invite users to explore and correct mental model deviations.

Ritter et al. (2014) emphasised the importance to support problem-solving and assist decision-making during user experience design. This can be achieved through user intention prediction with artificial intelligence, which is tested by Keyson et al. (2000) who developed a statistical model for thermostats to elevate human-machine interaction.

Conclusion

This paper illustrated why a user manual should not be included for an AI thermostat by demonstrating users' intention against using documentation instructions, the viability of removing user manuals for AI thermostats and providing superior approaches to support the user learning experience.

It has pointed out the common dissatisfaction towards documentation instructions caused by both the psychological nature of knowledge acquisition and the outdated designs that failed to address this nature and catch up with the new generation's expectations nurtured by modern technology. By analysing the characteristic of AI thermostats, with attribution to multiple technological traits that can easily sustain a better learning environment, the redundancy of it having a user manual booklet is testified.

As opposed to crusading for the devastation of user manuals, the paper is rather an appreciation of preceding designers' efforts to improve user experience through various forms of instructions, and an imagination of new forms of instructive learning empowered by emerging technologies. Printed user manuals should be regarded as a milestone in the development of user-centred design, and (hopefully) will soon be outshined by more user-friendly instruction systems.

Word count: 1869 / 1700

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